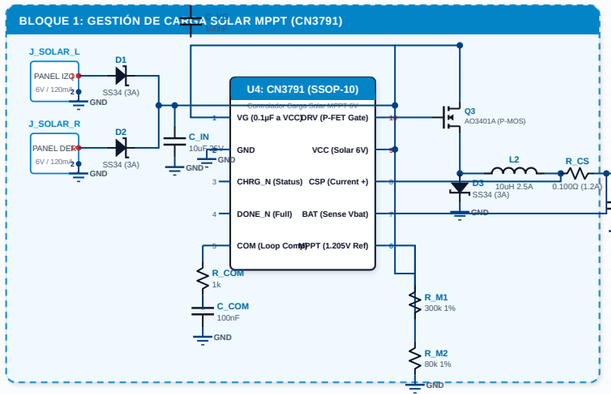
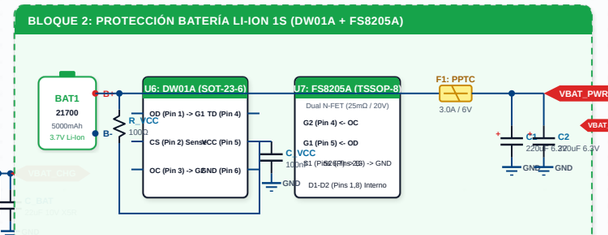
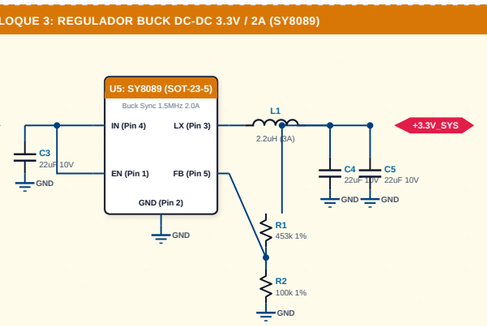
BLOQUE 1: GESTIÓN DE CARGA SOLAR MPPT (CN3791)



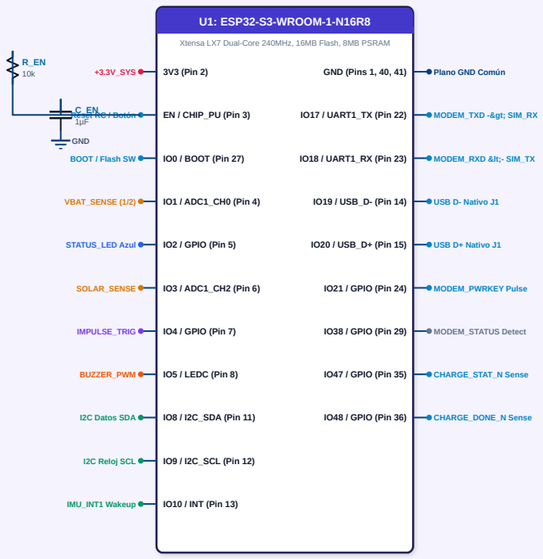
BLOQUE 2: PROTECCIÓN BATERÍA LI-ION 1S (DW01A + FS8205A)



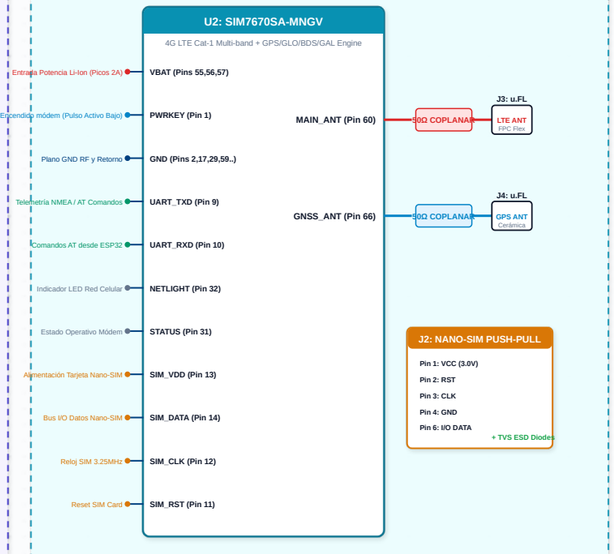
BLOQUE 3: REGULADOR BUCK DC-DC 3.3V / 2A (SY8089)



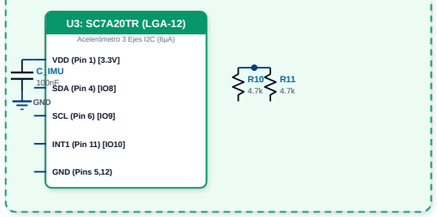
BLOQUE 4: MICROCONTROLADOR PRINCIPAL (ESP32-S3-WROOM-1-N16R8)



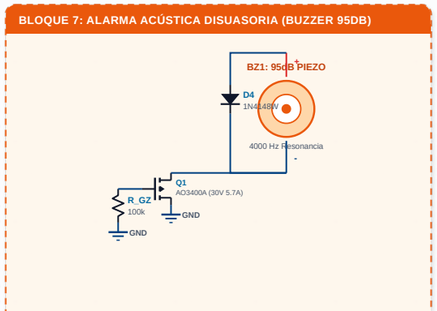
BLOQUE 5: TELEMETRÍA MÓVIL 4G LTE CAT-1 + GNSS (SIM7670SA)



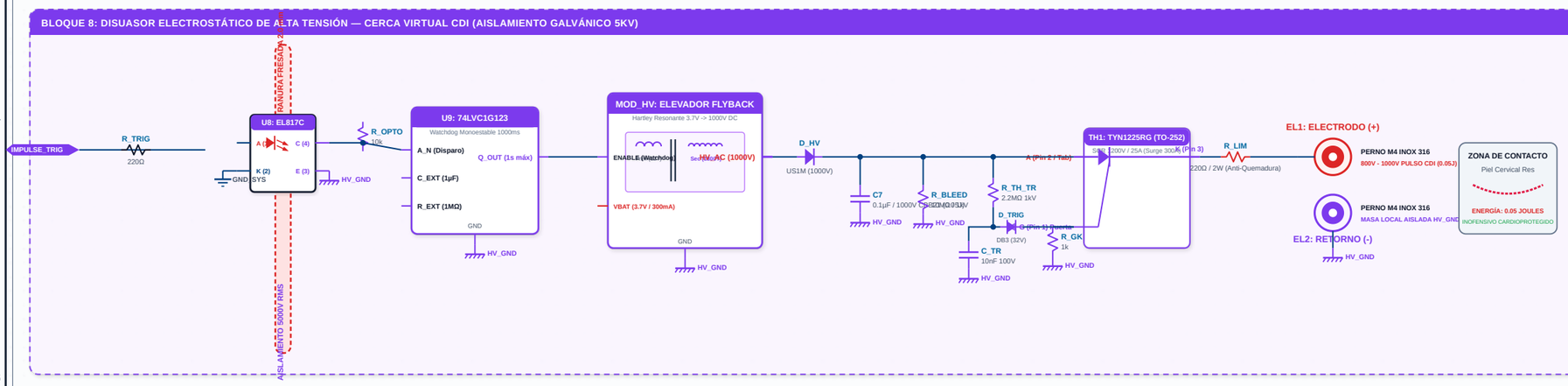
BLOQUE 6: SENSOR INERCIAL / PODÓMETRO GANADERO (SC7A20)



BLOQUE 7: ALARMA ACÚSTICA DISUASORIA (BUZZER 95DB)



BLOQUE 8: DISUASOR ELECTROSTÁTICO DE ALTA TENSIÓN — CERCA VIRTUAL CDI (AISLAMIENTO GALVÁNICO 5KV)

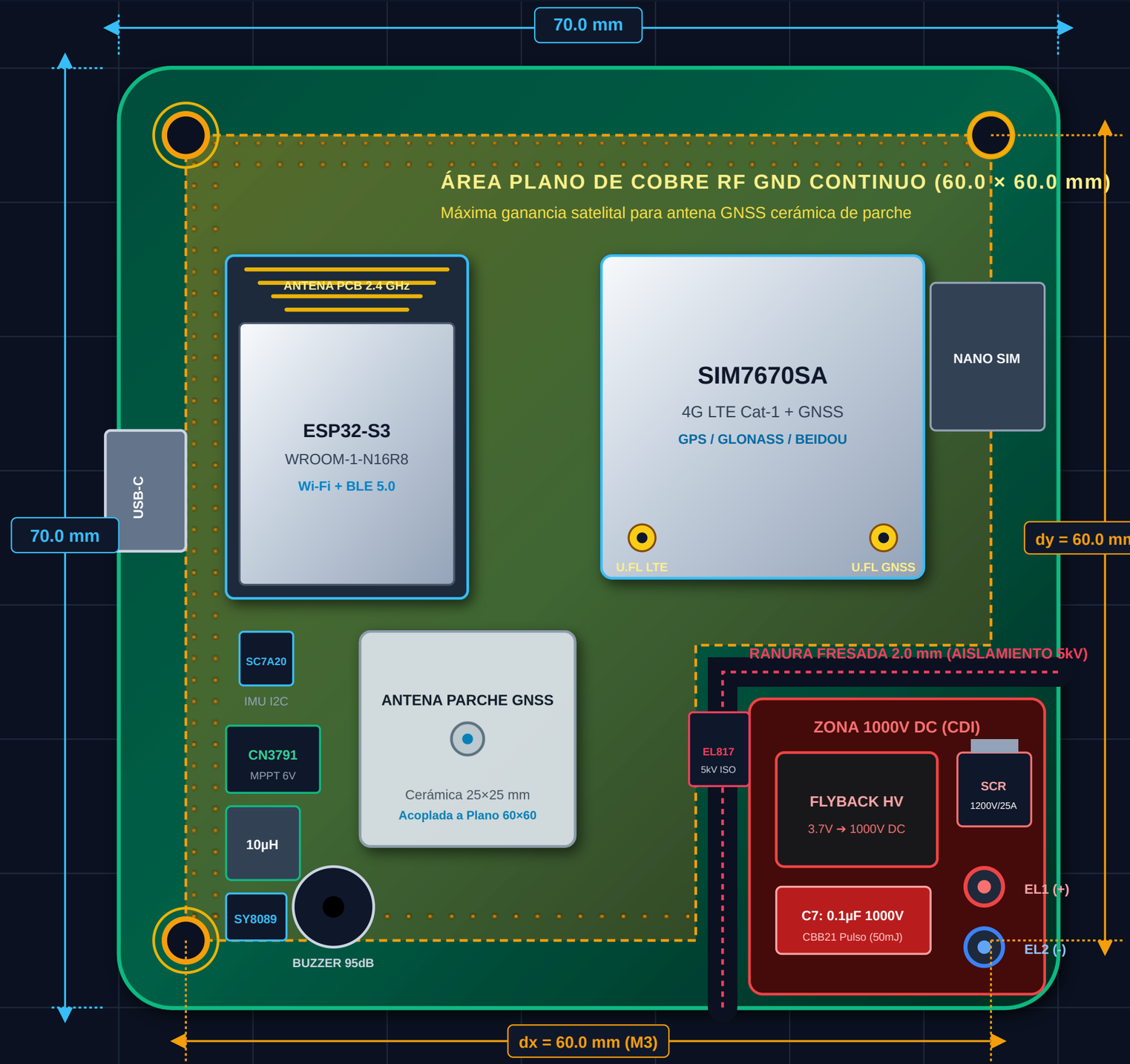


COW-AI TECHNOLOGIES S.A.	PROYECTO:	REVISIÓN:
DIAGRAMA DEL CIRCUITO COMPLETO ALL-IN-ONE (8 BLOQUES)	COLLARNET V2	2.2
FECHA: 2026-10-27 | DISEÑADOR: ING. HARDWARE CDMX	NORMA: IPC-2221 / IEC 60664-1 (5kV)	PLANO 1 DE 6

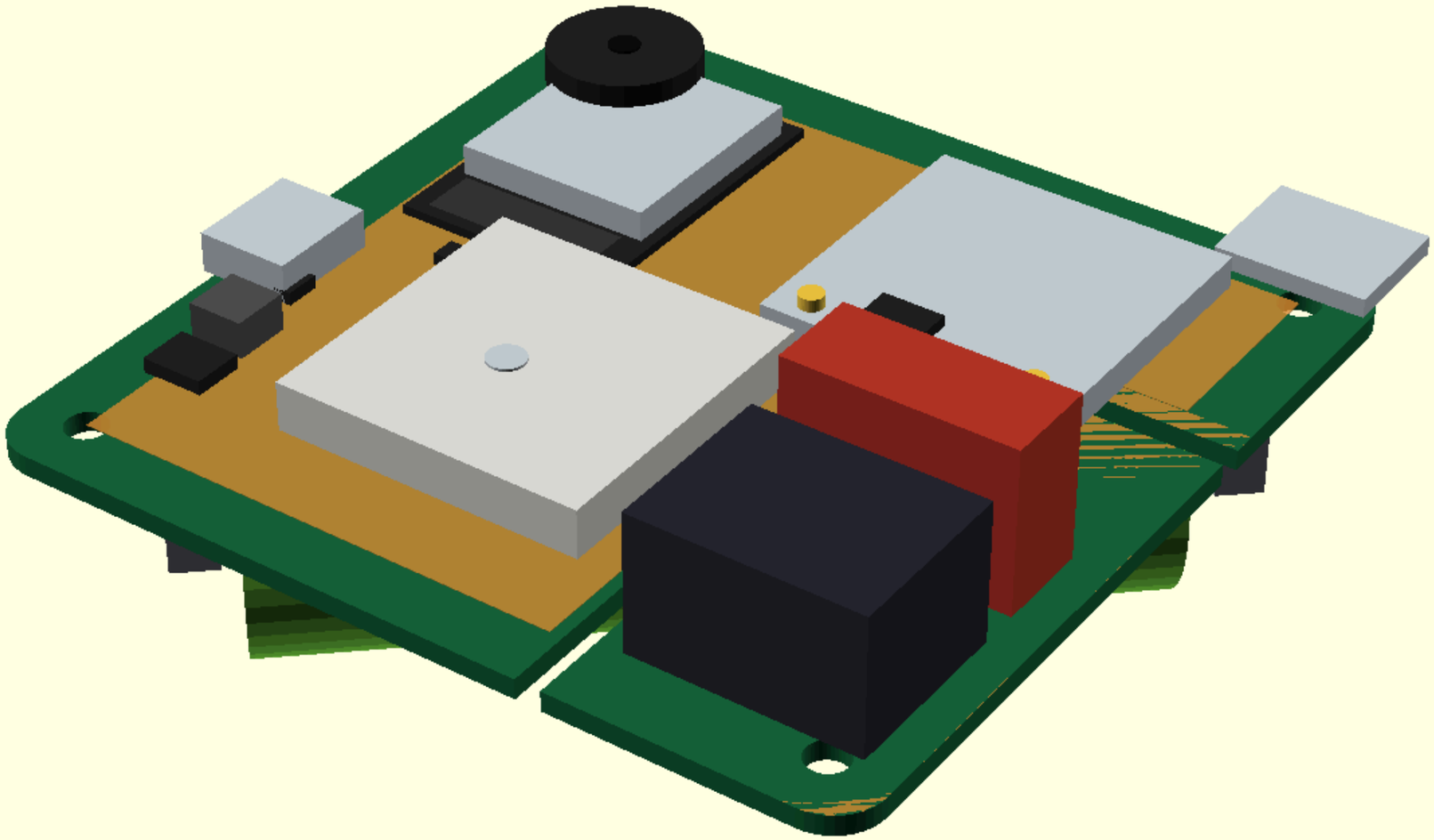
COLLARNET V3 — ARQUITECTURA CUADRADA GANADERA (70.0 × 70.0 mm)

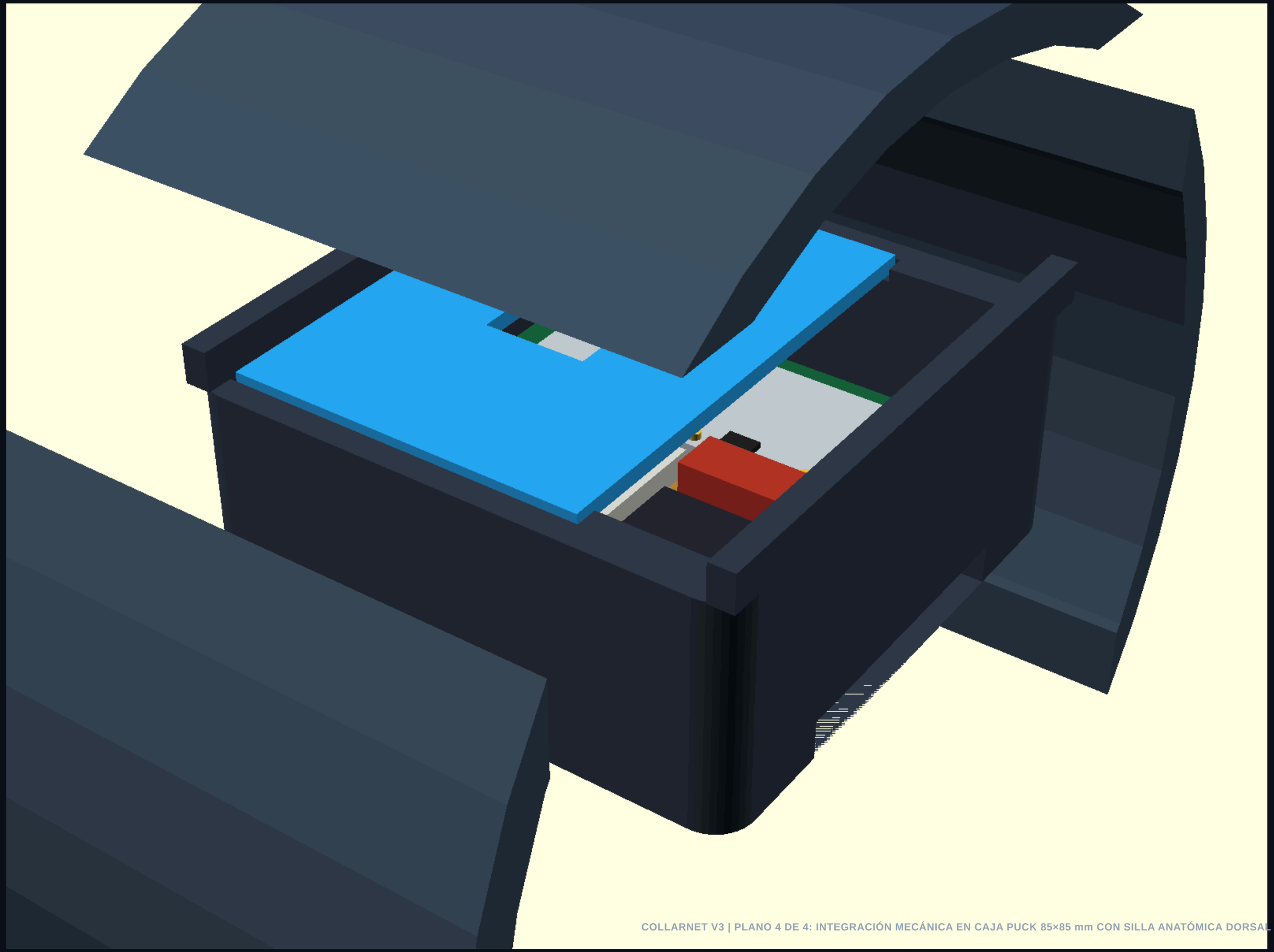
Plano de Cobre GNSS 60×60 mm | Batería Panasonic 18650 Diagonal | Aislamiento HV L-Slot 2.0 mm

ESCALA 10:1
100 px = 10.0 mm



Plano Cobre RF GND 60×60mm | Batería 18650 en Diagonal (Bottom) | Ranura Aislamiento 2mm (5kV) | Antena GNSS Parche 25×25mm





COLLARNET V3 | PLANO 4 DE 4: INTEGRACIÓN MECÁNICA EN CAJA PUCK 85×85 mm CON SILLA ANATÓMICA DORSAL